

Response Letter with Annotations

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Haroon, M., Wojcieszak, M., Chhabra, A., Liu, X., Mohapatra, P., & Shafiq, Z. (2023). Auditing YouTube's recommendation system for ideologically congenial, extreme, and problematic recommendations. *Proceedings of the National Academy of Sciences - PNAS*, 120(50), Article e2213020120. <https://doi.org/10.1073/pnas.2213020120>

Annotations

1. Purposes

- a. Motivation (one-sentence summary): The motivation of the study is to investigate whether YouTube recommendation algorithms systematically expose users to ideologically congenial content, especially deeper in the watch trail, and whether these content become progressively extreme and come from problematic channels.
- b. Specific Rationale (several bullet points)
 - ⑩ **Social concerns:** American society is becoming politically polarization, increasing hostility between partisans. The rise of social media platforms is one of the contributors of the radicalization.
 - ⑩ **Filter bubbles and rabbit holes:** YouTube's recommendation algorithm personalizes content based on users' ideology inferred from previous viewing behavior, which may lead to ideologically congenial recommendations. As users continue following recommendations, algorithmic personalization may intensify, especially for partisan users.
 - ⑩ **Inconclusive evidence:** Existing empirical evidence on YouTube algorithm is conflicting.
 - ⑩ **Methodological gap:** Existing analyses were based on untrained sock puppets without any watch history or real user data unable to isolate algorithm influence from user choice.
- c. Research Questions and/or Hypotheses
 - Identify IV, DV, mediator, moderator
 - ☞ **IV:** User's political ideology
 - ☞ **DV:** Ideological congeniality of recommendations, ideological extremity of recommendations, proportion of problematic channels
 - ☞ **Moderator:** position in recommendation trails(?)
 - Research questions:
 - ☞ **RQ1:** Are recommended videos congenial with users' ideology, especially deeper in the watch trail?
 - ☞ **RQ2:** Are recommendations deeper in the trail progressively more extreme, based on our classification of the ideological slant of each video?
 - ☞ **RQ3:** Do recommendations deeper in the trail direct users to conspiratorial, misinformative, or radical contents, based on extensive validated lists of problematic YouTube channels?

2. Method

- a. Research Design: Large-scale algorithmic audit using trained sock puppets
 - **Experiment:** 100,000 sock puppets divided into five ideological groups (Very-Left, Left, Center, Right, Very-Right).
- b. Sample (representative sample?)

Table 1. Data collection statistics for 100,000 sock puppets

	Training	Testing	Total
# of videos watched	9,930,110	5,393,820	15,323,930
# of unique videos	23,735	381,153	399,935
# of channels	1,256	119,811	120,073

- c. Measures
 - ⑩ **Ideological slant of videos:** Estimated on a scale from -1 (Very-Left) to +1 (Very-Right) by analyzing the Twitter followers of users who shared the video.
 - ⑩ **Ideological congeniality:** Measured by the regression coefficient between a sock puppet's trained ideology and the slant of recommended videos.
 - ⑩ **Ideological extremity:** Measured by how close the slant of recommended videos was to the extreme ends (-1 or +1).
 - ⑩ **Exposure to problematic channels:** Determined by checking if recommended videos came from channels on pre-defined lists of extreme, conspiratorial, or radical categories.
- d. Analysis: Comparative analysis across ideological groups and across depth in the recommendation trails.

3. Results and Conclusions

- a. Answers to RQs and Hypotheses
 - ⑩ **RQ1:** YouTube's algorithm recommends congenial content to its partisan users, although some moderate and cross-cutting exposure is possible and that congenial recommendations increase deeper in the trail for right-leaning users.
 - ⑩ **RQ2:** There were no meaningful increases in ideological extremity of recommendations deeper in the trail.
 - ⑩ **RQ3:** We show that a growing proportion of recommendations comes from channels categorized as problematic, with this increase being most pronounced among the very-right users.
- b. Include Key Figures/Tables

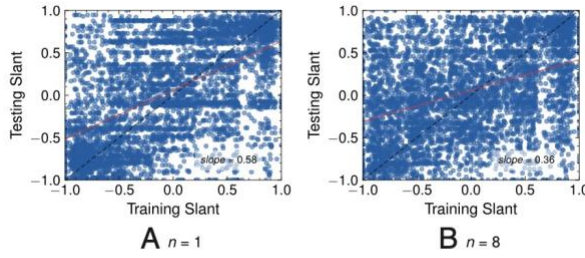


Fig. 1. Ideological congeniality between slant of training videos and the slant of top-1 ($n = 1$) and top-8 ($n = 8$) homepage recommendations. The higher the slope of the red line, the more congenial the recommendations to the training.

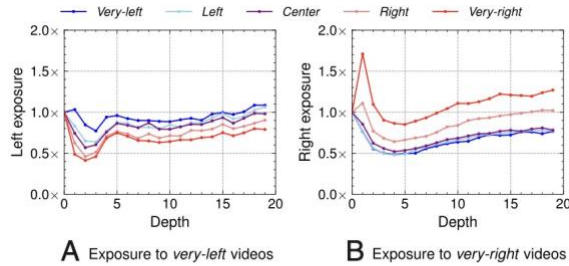


Fig. 4. Exposure to ideologically very-left and very-right videos in the up-next recommendations. The colored lines correspond to the ideology of the sock puppet. The x-axis corresponds to the depth d and the y-axis to the exposure E for the very-left and very-right ideologies. For example, in the right-hand figure at depth 1, the very-right sock puppet is exposed to 1.7 $\times$ more very-right videos.

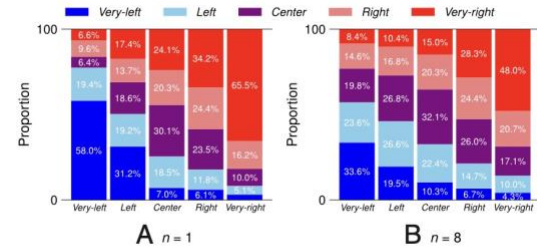


Fig. 3. The distribution of ideological categories in top-1 ($n = 1$) and top-8 ($n = 8$) homepage recommendations. For the very-left sock puppet, the top-1 recommendation is 58% likely to be congenial.

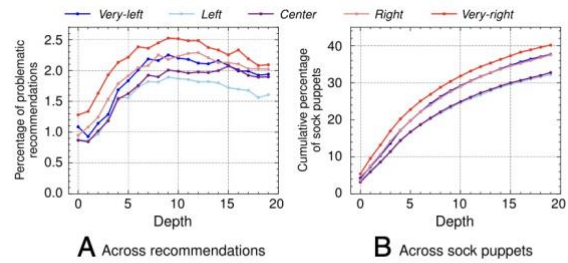


Fig. 6. Percentages of problematic video recommendations across (A) all recommendations and (B) across all sock puppets. The figure on the *Left* shows the percentage of problematic recommendations over all recommendations at each depth. The figure on the *Right* shows the cumulative percentage of sock puppets that encountered one or more problematic recommendations up to that depth.

c. Authors' Conclusions

4. Discussion

a. Implications

- ⑩ **Theoretical:** The study offers a holistic understanding of whether and how the algorithmic recommendation system of YouTube drives exposure to ideologically congenial and progressively more extreme and problematic videos.
- ⑩ **Methodological:** The study designs a sock puppet-based auditing methodology that allows us to systematically isolate the influence of the algorithm in recommendations to ideologically congenial and increasingly extreme and problematic content using trained sock puppets.

b. Limitations

- ⑩ The study did not include data on what real users click on and how they react to algorithmic recommendations and did not analyze attitudinal or behavioral outcomes of the tested recommendations.
- ⑩ The study was not designed to address questions such as how much does the algorithm change over time.
- ⑩ YouTube (and other platforms) does not function in isolation from an overarching information and communication ecology of its users. The study focuses on the algorithm, but “off-platform” consumption was not considered with regards to radicalization.

c. Future Directions

- ⑩ Future audits should attempt to account for how social media users interact with recommendations, and how individual and algorithmic factors together drive potential filter bubbles or algorithmic radicalization on platforms.
- ⑩ Studies extending the work to actual YouTube users and testing over time effects of algorithmic systems are needed.

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Response Letter (~600 words)

- Your assessment of the study

The study investigates whether YouTube algorithmic recommendations promote ideological congenial, extreme or problematic content to users, using 100,000 trained sock puppet accounts to examine how users' prior ideological viewing patterns shape homepage and up-next recommendations across different depths in the watch trail.

The study has made a significant methodological contribution by using algorithmic audits methodology incorporating traditional field experiments in social science to explore the algorithmic recommendation system of social media platforms, which functions as a "black box". The researchers used 100,000 trained sock puppet accounts that simulated users with different political ideologies, allowing them to examine algorithmic recommendation processes under controlled conditions. Compared to prior studies that rely on untrained bots or observed data from real users, this approach captures realistic viewing patterns while minimizing confounding influences of individual choices, which allows for causal inferences of how algorithmic recommendation systems work rather than users' personal viewing patterns.

Additionally, the study also advances traditional measures of ideological slant of channels which primarily rely on automatic analysis of video content, by using Twitter follower networks of users who share the video. Since "not all videos posted by a channel reflect that channel's ideology", this design captures the collective political ideology of the audiences of the videos, reflecting how the video content is socially interpreted rather than merely how it is linguistically framed.

Theoretically, the study contributes to the understanding of "filter bubbles" and "rabbit holes" of social media platforms. By distinguishing ideological reinforcement and radicalization, the study, shows that algorithmic recommendation systems primarily reinforce congenial exposure without necessarily driving users toward more extreme positions. It challenges the intuitive idea that algorithmic recommendation systems linearly push users toward political extremity by exposing them to congenial content aligning with their political ideology, making the society increasingly polarized.

- Weaknesses

The first weakness of the study is the lack of interaction patterns of these sock puppets beyond viewing videos. In this study, sock puppets did not like, comment, subscribe, share, or dislike content, which is commonly believed to influence recommendation outcomes. Excluding such interaction patterns, the study cannot capture a whole picture regarding how algorithmic recommendation systems work in the real world, potentially reducing the ecological validity.

Second, the measurement of users' political ideology was a bit simplified. Political ideology is not always single-dimensional. Instead, political ideology sometimes varies based on specific issues. For instance, individual may be against social issues such as

LGBTQ+ rights and immigration while supporting climate preservation and environmental protection. Therefore, the single-dimensional left-right measurement scale may not fully capture the complexity of individuals' political ideology.

The third weakness concerns the selection bias of shared videos. It is widely believed that videos with radical positions are more likely to be shared. In the study design, videos with niche topics or entertainment videos which did not meet the landmark threshold but still have evident political ideology were excluded, which might potentially create the sampling error.

- Other notes

Audit methodology is a very innovative method in computational social science which combines large-scale trained dataset with traditional field experiments, which helps explore phenomena in which the underlying mechanisms are opaque and complex. However, some platforms or websites have increasingly introduced bot-detection systems which can identify, restrict, or filter simulated users. Therefore, to what extent or under what conditions, using such simulated users may risk triggering the robot-detection system, which might prevent researchers from observing anything at all?